Generalising Human Fire:

Thoughts from using DAFI & LIFE for WHAM!

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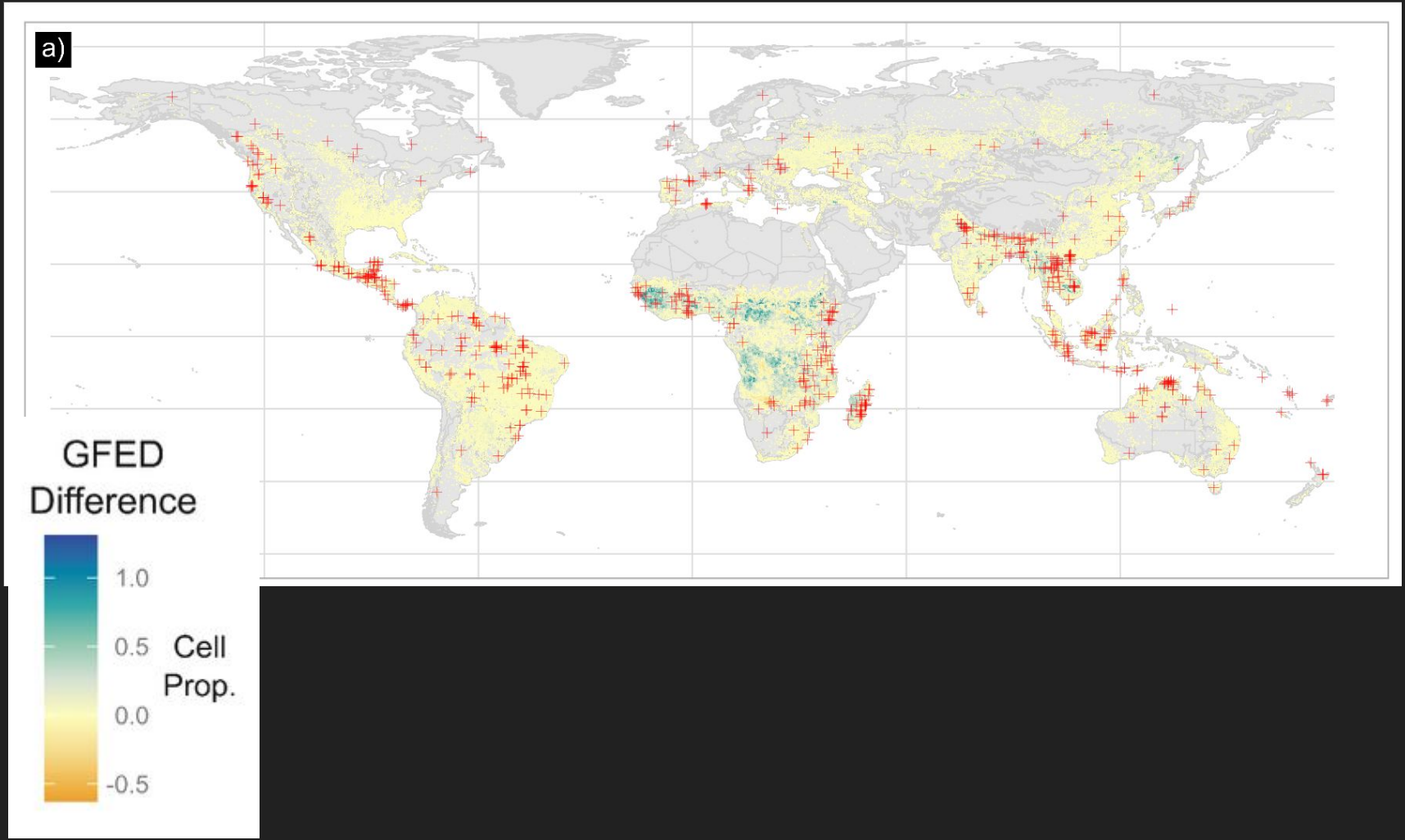
Databases of Case Studies (1990-2020)

	DAFI	**LIFE**
Data Points	1809 cases 504 sources	1708 practices 592 sources
Data Source	Literature: Primary, Secondary, Remote Sensing, Reviews	Literature: Primary
Fire Use Classes	*7 primary* [>20 total] Crop prep., Crop resid., Pasture, Hunt-Gather, Veg. clearing, Fuel mgmt, Arson	*8 primary* [29 total] Crops, Pasture, Hunt-Fish, Gather, Charcoal, Movement, Health, Social Signals
Policy and Governance	*Intervention:* Incentivise, Restrict, Ban *Rationale:* Environ., Econ., Health	*Intervention:* Regulations, Incentives, Voluntary *Intention:* Reduce, Retain, Grow, Introduce *Actor:* state, NGO, community

Building models from case studies

Human fire case study data are:

- Sparse: samples in space (and time)
- Non-uniform: inconsistent data collection btw. cases
- Varied: differences in study focus/emphasis
- Incomplete: not all case studies are fire-related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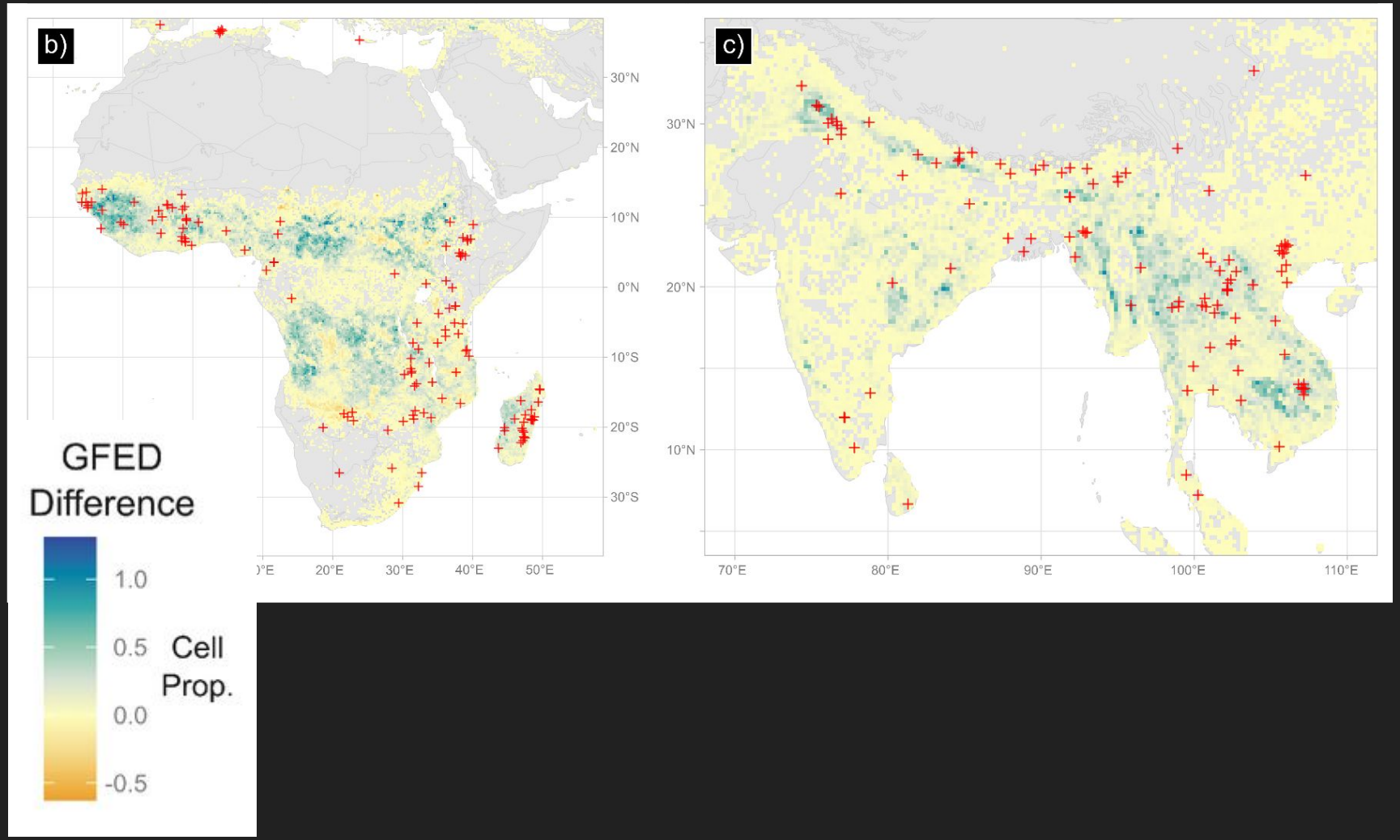
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A clear theoretical framing is needed to specify the categorisations, quantifications and decisions made when constructing a model from these data

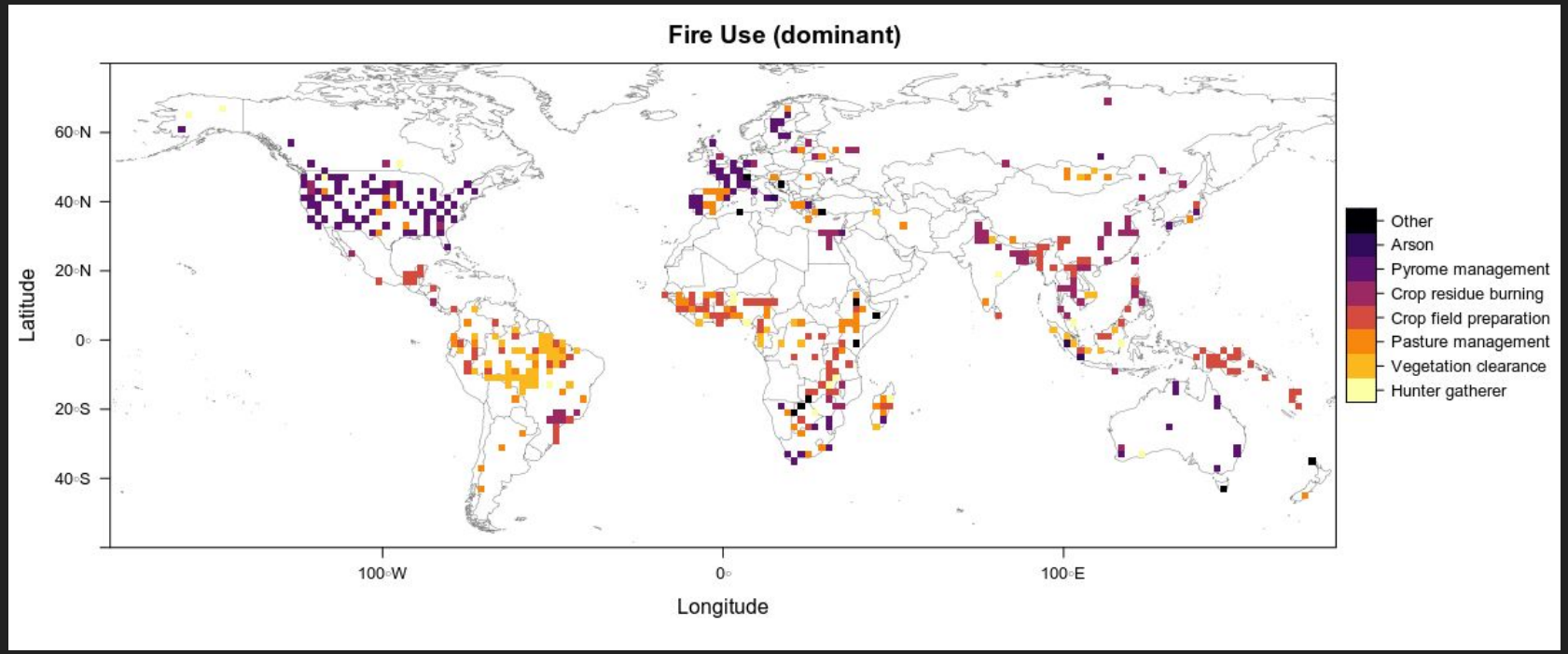
Fire Agents: Structuring a rational conception

	Non-extractive	Forest	Pasture	Cropland
Pre-Industrial	'Unoccupied'	Gatherer Hunter (S)	Pastoralist+ (S) Hunter Gatherer (S)	Swidden (S)
Transition	Urban resident	Forestry (small-scale, M) Agroforestry (M)	Farmer (extensive M|S) Small-holder (livestock-arable, M|S)	Small-holder (arable, S) Small-holder (arable, M)
Industrial	Urban resident Conservationist (fire exclusion) Tourist/recreation (M)	Logging (large-scale, M) Forestry (managed, M)	Farmer (intensive, M)	Farmer (intensive, M)
Post-Industrial	Conservationist (pyro-diversity) Tourist/recreation(M)	Abandoned (forestry)	Abandoned (pasture) Farmer (prescribed)	Abandoned (arable) Agroecologist

S: subsistence-oriented
M: market-oriented

DAFI Analysis

Seven fire use types describe 90% of the records



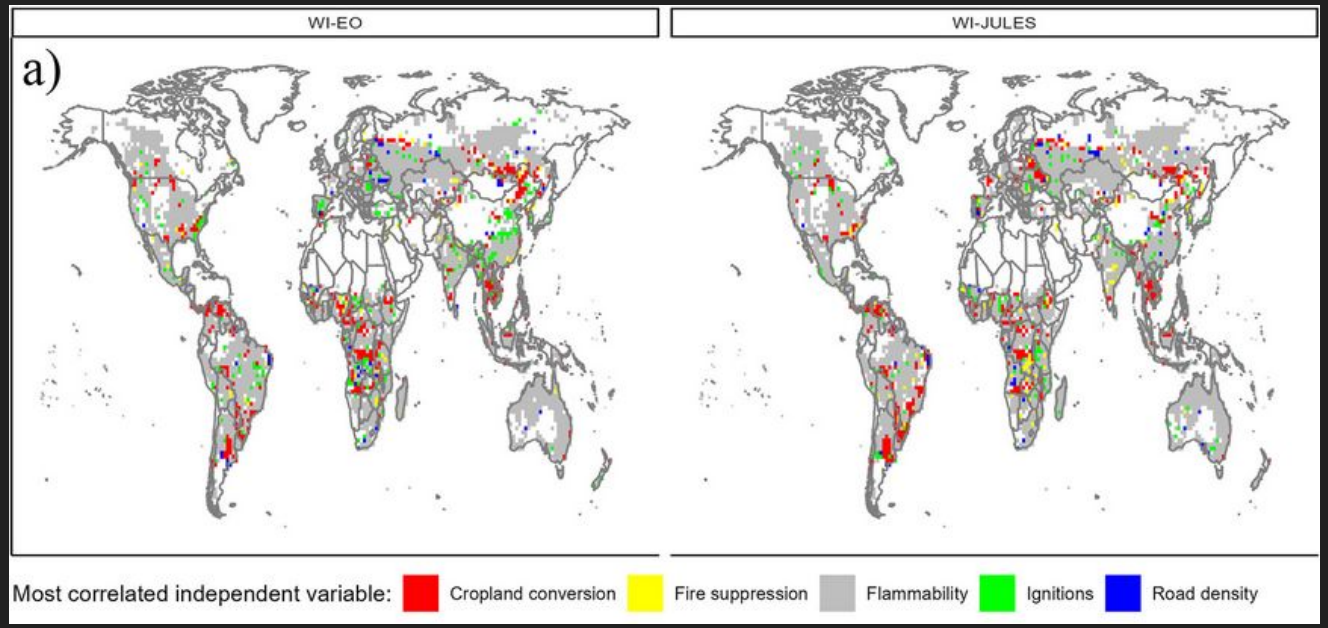
Millington *et al.* (2022)

Criticism of Generalisation

- Categorisation: Concerns about the use of categories like ‘hunter-gatherers’, ‘agriculturalists’ and ‘pastoralists’
- Context: fire is intricately place-based, nuanced, and driven by local climatic and ecological prompts - summarising many case studies masks this
- Complexity: the type of case study source (e.g. opinion vs direct observation) is vital for interpretation across case studies because expert opinion could reduce complexity (e.g. ‘slash and burn’ may have several purposes)

Coarse Computation vs Delicate Decisions

Our generalisation (WHAM!) improves modelling of *global patterns*, enables understanding general human drivers of changes in burning



Relationship of changes in unmanaged burned area [Fig 8a, Perkins *et al.* (2025)]

Coarse Computation vs Delicate Decisions

Challenges to informing local stakeholders:

- Context *is* important for making appropriate local decisions
- Huge grid cells in global models means that spatial heterogeneity within landscapes is masked (e.g. fuel connectivity)
- Large uncertainty in SSP projections underlying forecasts
- Annual burned area (product of ‘mean fire size’) is not necessarily a useful metric
    - We need to work towards fire size, intensity, rate of spread